

UST-TSILMA, Russian Federation

WMO: 234050

Lat: 65.4387N

Lon: 52.1974E

Elev: 260

StdP: 14.56

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.9	(c) -27.9	(d) -38.9	(e) 0.6	(f) -33.7	(g) -32.9	(h) 0.9	(i) -27.8	(j) 20.9	(k) 11.0	(l) 19.6	(m) 11.0	(n) 4.6	(o) 290	(p) 0.724

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	80.3	66.3	76.1	63.8	72.1	61.5	68.1	77.3	65.6	73.6	63.0	70.0	6.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	92.3	72.7	62.3	84.7	70.0	59.8	77.4	67.5	32.5	77.7	30.6	73.8	28.7	70.1	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	17.1	15.3	-36.1	85.0	8.1	3.9	-41.9	87.8	-46.6	90.0	-51.2	92.2	-57.0	95.0
			-36.1	70.6	7.8	3.0	-41.7	72.8	-46.3	74.5	-50.7	76.2	-56.3	78.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.5	2.6	4.0	17.5	29.3	40.9	52.7	59.7	53.6	45.0	32.7	17.3	9.5		
	DBStd	22.61	16.31	16.49	12.06	10.21	10.87	9.60	8.81	7.25	6.54	8.00	13.82	16.24		
	HDD50	7809	1469	1289	1009	623	330	82	15	42	177	538	981	1254		
	HDD65	12650	1934	1709	1474	1071	749	385	215	360	601	1002	1431	1719		
	CDD50	709	0	0	0	2	47	162	315	155	27	1	0	0		
	CDD65	74	0	0	0	0	1	16	49	8	0	0	0	0		
	CDH74	556	0	0	0	0	17	124	363	51	1	0	0	0		
	CDH80	104	0	0	0	0	1	21	75	7	0	0	0	0		
	WSAvg	7.6	7.8	7.7	8.0	8.0	8.4	7.9	7.0	6.8	6.8	7.7	7.4	7.6		
	PrecAvg	21.7	1.3	1.1	1.1	1.2	1.7	2.2	2.5	2.6	2.3	2.4	1.7	1.6		
(15) Precipitation	PrecMax	30.3	2.4	2.7	3.3	3.4	3.1	5.3	5.7	5.6	4.6	4.7	3.7	3.2		
	PrecMin	15.8	0.7	0.1	0.2	0.3	0.4	0.3	1.2	0.3	0.7	0.7	0.7	0.3		
	PrecStd	3.6	0.4	0.5	0.6	0.7	0.7	1.1	1.0	1.1	0.9	0.9	0.6	0.6		
	PrecStd	3.6	0.4	0.5	0.6	0.7	0.7	1.1	1.0	1.1	0.9	0.9	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.5	35.2	43.4	60.0	76.1	83.3	86.3	79.7	68.3	54.3	40.9	35.1		
		MCWB	31.1	33.8	37.4	48.4	59.4	65.8	69.1	67.3	58.6	49.2	39.4	33.3		
	2%	DB	29.5	31.9	37.2	50.2	70.0	77.6	82.0	73.6	61.9	49.2	37.0	33.3		
		MCWB	28.4	30.6	33.2	41.7	55.4	63.3	67.7	63.8	55.1	46.3	35.6	32.1		
	5%	DB	26.9	29.0	34.9	45.3	64.3	72.9	78.4	69.5	58.1	45.8	34.7	31.7		
		MCWB	25.9	27.6	32.1	38.3	52.4	60.2	65.9	61.4	53.4	43.7	33.5	30.7		
	10%	DB	23.6	25.9	32.5	42.0	58.8	68.3	74.3	65.6	54.7	43.2	32.9	29.3		
		MCWB	22.6	24.7	30.3	36.6	49.2	57.8	63.8	59.7	51.1	41.4	31.9	28.1		
	0.4%	WB	31.3	33.8	37.8	48.7	61.3	68.2	72.0	68.8	59.5	50.1	39.3	34.0		
		MCDB	32.1	35.0	42.5	58.3	72.8	79.2	81.8	77.4	66.4	52.7	40.4	34.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.5	30.6	34.1	42.4	57.2	64.7	69.7	65.6	56.4	46.5	35.9	32.4		
		MCDB	29.3	31.7	36.2	49.1	68.1	75.1	79.3	70.9	60.6	49.0	36.7	33.2		
	5%	WB	25.9	27.7	32.5	39.5	53.5	62.0	67.2	62.6	53.9	43.8	33.7	30.6		
		MCDB	26.9	28.9	34.4	43.8	62.4	70.3	76.1	67.4	57.4	45.6	34.5	31.7		
	10%	WB	22.7	24.6	30.4	37.3	49.7	58.7	64.9	60.2	51.5	41.5	32.0	28.2		
		MCDB	23.7	25.8	32.7	41.2	58.1	66.9	72.9	65.0	54.2	43.1	32.8	29.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	11.4	14.1	14.1	15.3	15.1	15.5	13.3	10.8	6.8	8.6	10.3		
		MCDBR	10.7	9.7	11.4	17.6	23.1	20.9	20.3	20.1	16.9	9.1	6.8	8.5		
		MCWBR	10.4	8.9	8.9	11.6	12.9	10.6	10.1	11.2	10.6	7.5	6.8	8.2		
		MCDBR	10.7	9.8	10.2	15.0	21.5	19.2	18.6	17.6	15.3	8.8	6.7	8.4		
	5% WB	MCWBR	10.5	9.3	8.6	10.6	12.7	10.9	10.0	10.8	10.6	7.6	6.8	8.2		
		MCWBR	10.5	9.3	8.6	10.6	12.7	10.9	10.0	10.8	10.6	7.6	6.8	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.216	0.270	0.305	0.338	0.344	0.354	0.411	0.371	0.329	0.292	0.214	0.214		
	taud		1.912	2.057	2.103	2.149	2.393	2.464	2.297	2.394	2.492	2.304	1.960	1.923		
	Ebn at Noon		77	186	238	259	273	272	249	250	238	184	75	14		
	Edh at Noon		7	19	30	36	32	30	35	29	21	15	6	2		
(44) All-Sky Solar Radiation	RadAvg		37	236	700	1195	1568	1614	1584	1048	549	196	50	9		
	RadStd		3	29	82	130	129	180	171	113	58	26	9	1		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page